

IS 1006 Fundamentals of Statistics Group
Project Proposal 2023

Group Details	Group number 17 Group members :- 1. S.M.D.L Wijerathne -17541 - 2023s20292 2. W.M.D.S Wijesuriya -17544 - 2023s20295 3. K.G.D.C.Wijesingha - 17542 - 2023s20293 4. A.H.M.K.Y.B.Wijekoon - 17540 - 2023s20291 5. K.K.K.D.Wijethilaka - 17545 - 2023s20296 6. M.B.I. Wijesiri - 17543 - 2023s20294
Title of the Project: Dark Side of Social Media	
Introduction: Social media can be defined as a digital way of communication and interaction, where it turns human lives into a global village. Nevertheless, their pervasive influence has introduced both positive and negative impacts, but highlighting the negative effects on individuals and societies. Social media sites and platforms like Facebook, WhatsApp, Instagram, TikTok, Youtube etc. have attracted millions of people in different age groups, especially targeting youths and adolescents all over the world in which many have integrated and addicted to using them in their daily lives. This dataset has been generated using synthetic data techniques with the help of NumPy and pandas. The data is artificially created to simulate real-world social media usage patterns. We hope to analyze this data and uncover the negative impacts of social media in human lives.	

Objectives:

The key objective of analyzing the dark side of social media data set is to understand and diminish the negative impacts it has on mental health, social behavior, and information integrity. By investigating issues such as social isolation, addiction, decrease in privacy and effects on well-being, this analysis focuses on to inform the development of effective interventions, policies, and educational initiatives that promote safer and more responsible use of social media platforms.

Data:

Qualitative Variables: - Gender, Location, Debt, Own property, Profession, Demographic, Platform, Video Category, Frequency, Watch Reason, Device type, Operating System(OS), Current activity, Connection type, Watch Time.

Quantitative Variables: - Age, Income, Total time spend, Number of sessions, Video length, Engagement, Important Score, Time spent on video, Number of videos watched, Scroll rate, Productivity loss, Satisfaction, Self control, Addiction level .

Total Variables - 29

Number of Observations - 1000

Proposed Analysis Plan and Methodology:

As mentioned in the data set, we are using all 15 categorical variables and 14 quantitative variables for the analysis. Pie charts and bar charts will be used to show categorical data, and time plots, box plots, scatter plots, and histograms will be used to examine quantitative data. We thought to obtain the mean, median, standard deviation, quartiles and regression lines to get essential numerical summaries. To construct the graphs and the data sheet required to create the report, we use the Mini-tab software. At last, we hope to analyze the data set effectively using all these methodologies.

References:

- <https://www.kaggle.com/datasets/muhammadroshaanriaz/time-wasters-on-social-media/data>
- <https://www.minitab.com/en-us/products/minitab/>
- https://www.amazon.com/Dark-Side-Social-Media-Psychological/dp/0128159170/ref=sr_1_1?crid=29IBSOOXAK719&dib=eyJ2IjoiMSJ9.vLXvHajRzdUrt6ifQSnhdIhtIIvBYJEeA6bapBgrLC5Cksj3ZmFDDYAQrm_B82XvalbDuUcxohQjLo-_Zuf-NTDedSZ0yimQQiGxaBu0cSi-wuf0SYAw8SJb42EKtR6u8JcnehZ-OjgHz2AGs-2PZp-ANkKPv2LAZJamORiN_9MtE8nII28ODS2Koi0wZvAziK_3hjVyLNxwI0ReUnaLGRVUVO5PBw8iM2Pwc6H8Ru4.QE2_ykR7qI1MlqcEudtQBx8R7dNa4yUo_WSpwA3_Ugg&dib_tag=se&keywords=dark+side+of+social+media&qid=1722517310&s=books&sprefix=dark+side+of+social++%2Cstripbooks-intl-ship%2C410&sr=1-1&asin=0128159170&revisionId=&format=4&depth=1